

HAZEM MASARANI

Data Scientist · Applied Machine Learning · Analytics

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GitHub · Kaggle · ResearchGate · LinkedIn

PROFESSIONAL SUMMARY

Data scientist and published ML-systems researcher experienced in statistical modeling, machine learning, data engineering, and decision-focused analytics. Built end-to-end Python/SQL pipelines and analysis products for telecom operations; pursuing an M.S. in Data Science with a 3.96 GPA. **ANALYTICAL TOOLKIT**

Methods: Regression, classification, clustering, feature engineering, cross-validation, BIC model selection, EDA

Languages/Libraries: Python, SQL, R, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch

Data/Visualization: ETL, PostgreSQL, MySQL, SQL Server, MongoDB, Power BI, Plotly, Matplotlib, FRED **EXPERIENCE**

& RESEARCH

PACELab, Stony Brook University

Graduate Researcher | Advisor: Prof. Anshul Gandhi

Aug 2025–Present

Stony Brook, NY

- Design multi-GPU ML experiments and evaluate throughput, latency, memory, communication overhead, and numerical consistency.
- Research scalable inference for selective state-space models and validate tensor-parallel implementations against reference outputs.
- Co-authored a published preprint introducing state caching, locality-preserving sharding, and quantized collective communication.

e&

Senior Data Automation & Network Data Engineer

Jan 2024–Sep 2024

Egypt | Hybrid

- Led Python/SQL pipelines and analytics integrating SNMP traps, outages, topology, rollout, availability, power, and maintenance data.
- Built root-cause logic linking service outages to network topology and developed CapEx/OpEx impact algorithms to prioritize site repairs.
- Created Power BI/Plotly monitoring products and geospatial views supporting NOC, field-maintenance, power, and investment decisions.

e&

Junior Data Automation & Network Data Engineer

Jul 2023–Jan 2024

Egypt | Hybrid

- Integrated multi-source data into the eSW analytics platform and developed a chatbot to streamline data access and analysis.

SELECTED DATA PROJECTS

Federal Funds Rate Prediction | [GitHub](#)

End-to-end FRED analysis with preprocessing, inflation-regime features, regression, clustering, BIC, and cross-validation.

Spaceship Titanic Classification | [GitHub](#)

Kaggle workflow covering missing-data treatment, feature engineering, model comparison, and passenger-outcome prediction.

FM-Index Engine | [GitHub](#)

Computational data pipeline implementing suffix arrays, BWT, compressed indexing, and efficient substring queries. **PUBLI-**

CATION

Scaling State-Space Models on Multiple GPUs with Tensor Parallelism | [arXiv:2602.21144](#)

Dutt, Shah, Masarani, Gandhi · February 2026 **EDUCATION**

Stony Brook University — M.S. Data Science, **GPA: 3.96/4.00**

Expected Dec 2026

Alexandria University — B.Sc. Computer & Communication Engineering, Honors, GPA: 3.47/4.00

2022